



ERADICATION OF POLIO & PAKISTAN



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Fighting polio in Pakistan

Instead of insisting on the oral polio vaccine, using the inactivated polio vaccine along with other vaccines will help



T. JACOB JOHN & DEVIANTA SHARADAPALAN

Last month, the polio eradication programme in Pakistan was in the news for all the wrong reasons. On April 22, a government hospital in Muzaffargarh in Punjab district was set on fire after many children allegedly fell sick after being given the anti-polio vaccine. On April 23 and 24, in two separate incidents, two police officers guarding vaccination centres were shot dead. On April 25, in Chaman, which borders Afghanistan, a police worker was shot dead and his helper injured. Since December 2012, nearly 90 people have been killed in the country for working to eradicate polio. Due to recurrent threats to workers, the Pakistan government has now suspended the anti-polio drive.

Cases of wild poliovirus type 1

This is the worst time to take this decision. This year alone, eight paralysed children with wild poliovirus type 1 (WPV1) have been found in Pakistan. Environmental surveillance by testing sewage samples has shown 34 WPV1-positive samples, in the provinces of Khyber Pakhtunkhwa, Balochistan, Punjab and Sindh. In the past week alone, 12 sewage samples were found to be positive for WPV1.

This is a worrying sign. With suspended immunisation activities, WPV1 will spread fast and the number of polio cases could increase and cause an outbreak. If Pakistan cannot eliminate polio, the global eradication programme is sure to stall.

When India eliminated WPV1 in January 2011, the Global Polio Eradication Initiative (GPEI), a public-private partnership led by national governments with five partners, did not ask if Pakistan would be able to follow suit. It simply assumed it would. This was unrealistic. In Uttar Pradesh and Bihar, oral polio vaccine (OPV) coverage of 90-95% was maintained with an average of 15 doses per child from 2011. There was full cooperation from the health workers and the public. The war on polio requires such intensity and coverage and it is unrealistic to expect this in Pakistan, where polio eradication is falsely depicted as a Western agenda with the sinister motive of reducing fertility.

The GPEI has pinned all its hopes on the

OPV and has excluded the alternate inactivated polio vaccine (IPV) to eradicate WPV1. The OPV is cheap and easy to give to children, but it has to be given to them again and again in pulse campaigns since its efficacy is poor. On the other hand, the IPV is highly effective and needs to be given just two-three times as part of routine immunisation.

Risk of polio outbreaks

The OPV has another problem. If coverage declines (as is bound to happen in Pakistan), vaccine viruses will spread to children who are not vaccinated, back-mutate, de-attenuate and become virulent. Such viruses are called circulating vaccine-derived polioviruses (cVDPV). They can cause polio outbreaks. Thus Pakistan will soon be at risk of polio outbreaks by both WPV1 and cVDPV.

It is to avoid the emergence of cVDPV that India strives to maintain high OPV coverage through routine immunisation, Mission Indradhanush and annual national pulse campaigns. In 2008, Papua New Guinea developed a cVDPV polio outbreak as OPV coverage fell to 60%. In 2013, as OPV coverage fell to 57%, Syria had an outbreak of cVDPV polio.

There is yet another problem in Pakistan. With the OPV being identified as the weapon in the war on polio and with some in Pakistan believing that the aim of eradication is to reduce fertility, a vaccine is given only three or four times, not 15-20 times.

Hope is not lost for polio eradication provided that the GPEI reverts on its insistence on the OPV and uses the IPV along with other common vaccines. IPV-containing vaccines could be included in the routine immunisation programme and given without attracting the attention of militants. The false propaganda about polio vaccination in Pakistan will then lose its sting. While near 100% coverage with the OPV is necessary, 85-90% coverage with the IPV given in a routine schedule would be sufficient.

If the GPEI insists on the OPV as the only weapon against polio, we have hit the end of the road in Pakistan. But the world cannot afford to lose this war on polio. India could show the way forward by giving the IPV in its universal immunisation programme (at least two doses and preferably three) and then discontinuing the infectious OPV altogether.

T. Jacob John, a retired professor of Clinical Virology, taught at the Christian Medical College, Vellore, and Shreeya Sharadapalan is a Postgraduate in Microbiology, a Specialist in Paediatric Hepatology, and a Consultant in Paediatric Hepatology, New Mumbai.

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UPSC CSE – GS III

GS III ... Science and Technology

Science and Technology - developments and their applications and effects in everyday life
Achievements of Indians in science & technology; indigenization of technology and developing new technology.

3 Fighting polio in Pakistan

Instead of insisting on the oral polio vaccine, using the inactivated polio vaccine along with other vaccines would help. Last month, the polio eradication programme in Pakistan was in the news for all the wrong reasons. On April 22, a government hospital in Mashokhel in Peshawar district was set on fire after many children allegedly fell sick after being given the anti-polio vaccine. On April 23 and 24, in two separate incidents, two police officers guarding vaccinators were shot dead. On April 25, in Chaman, which borders Afghanistan, a polio worker was shot dead and her helper injured. Since December 2012, nearly 90 people have been killed in the country for working to eradicate polio. Due to recurrent threats to workers, the Pakistan government has now suspended the anti-polio drive.

Cases of wild poliovirus type 1

This is the worst time to take this decision. This year alone, eight paralysed children with wild poliovirus type 1 (WPV1) have been found in Pakistan. Environmental surveillance by testing sewage samples has shown 91 WPV1-positive samples, in the provinces of Khyber Pakhtunkhwa, Balochistan, Punjab and Sindh. In the past week alone, 13 sewage samples were found to be positive for WPV1.

This is a worrying sign. With suspended immunisation activities, WPV1 will spread fast and the number of polio cases could increase and cause an outbreak. If Pakistan cannot eliminate polio, the global eradication programme is sure to stall. When India eliminated WPV1 in January 2011, the Global Polio Eradication Initiative (GPEI), a public-private partnership led by national governments with five partners, did not ask if Pakistan would be able to follow suit; it simply assumed it would. This was unrealistic. In Uttar Pradesh and Bihar, oral polio vaccine (OPV) coverage of 98-99% was sustained with an average of 15 doses per child from 2003. There was full cooperation from the health workers and the public. The war on polio requires such intensity and coverage and it is unrealistic to expect this in Pakistan, where polio eradication is falsely depicted as a Western agenda with the sinister motive of reducing fertility.

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The GPEI has pinned all its hopes on the OPV and has excluded the alternate inactivated polio vaccine (IPV) to eradicate WPVs. The OPV is cheap and easy to give to children, but it has to be given to them again and again in pulse campaigns since its efficacy is poor. On the other hand, the IPV is highly efficacious and needs to be given just two-three times as part of routine immunisation.

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It is to avoid the emergence of cVDPV that India strives to maintain high OPV coverage through routine immunisation, Mission Indradhanush and annual national pulse campaigns. In 2018, Papua New Guinea developed a cVDPV polio outbreak as OPV coverage fell to 60%. In 2017, as OPV coverage fell to 53%, Syria had an outbreak of cVDPV polio. There is yet another problem in Pakistan. With the OPV being identified as the weapon in the war on polio and with some in Pakistan believing that the aim of eradication is to reduce fertility, a vaccine is given only three or four times, not 15-20 times.

Hope is not lost for polio eradication provided that the GPEI relents on its insistence on the OPV and uses the IPV along with other common vaccines. IPV-containing vaccines could be included in the routine immunisation programme and given without attracting the attention of militants. The false propaganda about polio vaccination in Pakistan will then lose its sting. While near-100% coverage with the OPV is necessary, 85-90% coverage with the IPV given in a routine schedule would be sufficient.

...Contd.



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T. Jacob John, a retired professor of Clinical Virology, taught at the Christian Medical College, Vellore, and Dhanya Dharmapalan is a Paediatric Infectious Disease Specialist in Apollo Hospitals, Navi Mumbai

PREVIOUS EDITORIAL DISCUSSION ON 12th NOVEMBER, 2018

Protecting against polio

Why the inactivated polio vaccine is essential for India



S. PRASAD

With wild polio virus strains reduced by 99.9% since 1988, the world is inching towards eradicating polio. But unfortunately, more children today are affected by the live, weakened virus contained in the oral polio vaccine (OPV) that is meant to protect them. The weakened virus in the vaccine can circulate in the environment, occasionally turn neurovirulent and cause vaccine-derived poliovirus (VDPV) in unprotected children. While the wild-type virus has caused 22 and 25 polio cases in 2017 and 2018 (as on October 30, 2018), respectively, in just two countries (Pakistan and Afghanistan), VDPV was responsible for 96 and 79 polio cases in more countries during the same periods. "Paradoxically, vaccination (using OPV) has become the main source of polio paralysis in the world," notes a 2018 paper in *The Lancet*.

The VAPP burden

While circulating VDPV strains are tracked, and outbreaks and cases are recorded and shared, little is known about vaccine-associated paralytic poliomyelitis (VAPP) cases, particularly in India. VAPP occurs when the virus nerve virulent within the body of a recently vaccinated child and causes polio. The frequency of VAPP cases varies across countries. With high-income countries switching to the inactivated polio vaccine (IPV) that uses dead virus to immunise children, the VAPP burden is concentrated in low-income countries which continue to use the OPV.

In spite of the World Health Organisation asking all countries using the OPV to include a "continuous and effective system of surveillance" to monitor the frequency of VAPP in 1982, India did not comply. Data on VAPP became available only years after active polio surveillance was initiated in 1997, says Jacob John, a virologist and formerly with the Christian Medical College, Vellore, and a polio expert, and Shanya Dharmapalan in a paper published in September in



"While high-income countries preferred the IPV, India and other low-income countries continued to rely on the OPV." A child being administered the polio vaccine in Kolkata. — AP

the Indian Journal of Medical Ethics. However, even after 1997, India did not count VAPP cases. "This is because it does not add value to the polio elimination programme," says Pradeep Halder, Deputy Commissioner of the Immunisation Division, Ministry of Health and Family Welfare.

The justification that VAPP cases can be ignored as they are "sporadic and pose little or no threat to others" is ethically flawed. The stand that VAPP cases are epidemiologically irrelevant is ethically problematic, note Dr. John and Vipin M. Yashibhatha in a 2012 paper in *Indian Pediatrics*.

Many member countries autonomously chose the IPV over the OPV, mainly to avoid any risk of VAPP. In India, the VAPP cases can be avoided once the government stops using the OPV to immunise children. "India ignored the problem of VAPP until their numbers were counted," writes Dr. John. A paper and a letter published in 2002 in the *Bulletin of the WHO* said the number of VAPP cases in India in 1998, 2000 and 2001 were 181, 129 and 109, respectively.

The WHO had suggested a rate of 1 case of VAPP per million births and had estimated the annual global burden of VAPP to be approximately 120 cases in 2002. Under those circumstances, India's share would have been merely 25 VAPP cases per year, based on the annual birth cohort of 25 million. But the observed number of cases in India in 1999 was 181. "This indicates that the actual risk is seven

times the expected number... It is reasonable to assume that there would be 400-600 annual cases of VAPP globally," Dr. John wrote in 2002 in the *Bulletin*. That would have meant that there were 100-200 VAPP cases in India each year. The global estimated incidence of VAPP was then revised to 200-400 cases.

Despite knowing that there is a higher burden of polio caused by wild viruses, India continued to use the OPV. "The decision to use only the OPV was faulty. Parents were obliged to accept the OPV and face the consequences of VAPP as well as VDPV," Dr. John says.

Says Dr. Halder, "India's goal was to eradicate polio, and the OPV was crucial for that. The IPV produces humoral immunity (involving antibodies in body fluids) so the immunised child does not get paralysis, but it can't stop the circulation of wild polio viruses. For instance, no polio cases were seen in Israel but wild polio viruses were detected in the environment. The viruses will continue to circulate in the community."

Dr. John counters this. "The primary objective of polio vaccination is to prevent the disease, which the OPV failed to fully achieve. The OPV was used for eradication purposes but without fully protecting the children. When you give a vaccine, you must ensure that the child doesn't get polio. Only the IPV can do that. A child has to be given several doses of the OPV. Even then, the OPV doesn't fully protect the child. There was no reason for not using both the IPV and

the OPV."

It is easier to administer the OPV than the IPV and the cost per dose of OPV is also lower than that of the IPV. However, the OPV failed poorly on two important counts: safety and efficacy. "Administering the OPV was easier than the IPV but no cost-benefit analysis was done before choosing the OPV," says Dr. John. "Three doses protected only two-thirds of Indian children and many developed polio before they turned one year. So we had to give more doses per child."

While high-income countries preferred the IPV, India and other low-income countries continued to rely on the OPV. India licensed the IPV only in 2006 but did not introduce it in routine immunisation.

Switching to IPV

"The reason for not switching over to the IPV is because global production was too low to meet India's demand. India is the largest cohort. It needs 48 million doses per year to immunise all children," Dr. Halder says.

This is a hefty excuse. As Padma Bhargava noted in an article in *The Hindu* (2008), the decision to manufacture the IPV in India was taken in 1986 and a company was eventually set up with technology transfer from France. The minutes of the meeting that year in Delhi read: "Indigenous production of IPV before 1991 shall be aimed at... As new IPV programme ramps up, the OPV will ramp down." But the plan was shelved.

The IPV is essential for post wild-type polio virus eradication, to get rid of VDPV and VAPP. The globally synchronised switch from trivalent to bivalent OPV in mid-2016 was accompanied by administering a single dose of the IPV prior to administering the OPV. "A single dose of the IPV given before the OPV prevents VAPP cases," Dr. John says. A single dose of the IPV primes the immune system and the antibodies against the polio virus, seen in more than 90% of immunised infants, notes a paper in *The Lancet*.

With no way of monitoring VAPP cases in India, there is no way of knowing if the use of a single dose of IPV followed by immunisation using bivalent OPV has led to a reduction in the number of VAPP cases.

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EDITORIAL DISCUSSION

UPSC CSE – GS III

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Science and Technology - developments and their applications and effects in everyday life
Achievements of Indians in science & technology; indigenization of technology and developing new technology.

IMPORTANT QUESTION

ESSAY/ MAINS / GD / INTERVIEWS

Examine the need to switch-over to Inactivated Polio Vaccine, with specific reference to India.



UKRAINE

KAZAKHSTAN

UZBEKISTAN

KYRGYZSTAN

TAJIKISTAN

AFGHANISTAN

PAKISTAN

CHINA

AFGHANISTAN
(MA)

Bay of Bengal

- Pakistan Government suspended the anti-polio drive following increasing number of attacks on polio workers.
- Countrywide campaign to administer anti-polio drops to 39 million children under five years of age was launched.
- However, the campaign, ran into trouble after reports that several children were taken to hospital as they fell sick.
- Pakistan is one of the three countries, where polio is still endemic.
- Since December, 2012, nearly 90 people have been killed in the country for working to eradicate polio.

Pakistan and Afghanistan, along with Nigeria, are the only three countries in the World, where polio remains endemic.



ORAL POLIO VACCINE

- 1 It contains live weakened virus.
- 2 OPV was used for eradicating purposes, but, without fully protecting the children.
- 3 It is easy to administer OPV than IPV.
- 4 Cost per dose of OPV is also lower than that of IPV.
- 5 India and low-income countries continued to rely on the OPV.

INACTIVATED POLIO VACCINE

- 1 It uses dead virus to immunize the children.
- 2 IPV produces humoral immunity involving antibodies in body fluids, so the immunised child does not get paralysis, but it can't stop the circulation of wild polio viruses.
- 3 The viruses will continue to circulate in the community.
- 4 High-income countries preferred IPV.
- 5 India licensed the IPV only in 2006 but, did not introduce it in routine immunisation.

PROBLEMS WITH LACK OF ORAL POLIO VACCINE IMMUNISATION AND PROBLEMS



- 1** It is cheap and easy to administer to children, it has to be given again and again in pulse campaigns since its efficacy is poor.
- 2** If coverage declines, vaccine viruses will spread to children, who are not vaccinated. It will back-mutate and becomes virulent.
- 3** Such viruses are called circulating Vaccine-Derived Polioviruses (cVDPV). This can cause polio outbreaks.
- 4** Pakistan will soon be at the risk of polio outbreaks by both WPV1 and cVDPV.

MORE ON CIRCULATING VACCINE DERIVED POLIOVIRUSES



Be Wise!
Get your child
fully immunized

- 1** In 2018, Papua New Guinea developed a cVDPV polio outbreak as OPV coverage fell to 60%.
- 2** In 2017, as OPV coverage fell to 53%, Syria had an outbreak of cVDPV polio.
- 3** That is the reason India strives to maintain high OPV coverage through routine immunisation i.e., Mission Indradhanush and Annual National Pulse Campaigns.

THE NEED TO INTRODUCE INACTIVATED POLIO VACCINE



1 Inactivated Polio Vaccines are highly efficacious, they need to be given just two-three times as part of routine immunisation.

2 IPV can be used along with other common vaccines.

3 IPV containing vaccines could be included in the routine immunisation programme, they can be given without attracting the attention of militants.

4 The false propaganda about polio vaccination will then vanish.

5 100% coverage with OPV is necessary, but, 85-90% coverage with IPV would be sufficient to prevent outbreaks.

India should also consider giving IPV in its Universal Immunisation Programme and then discontinuing the infectious OPV altogether.

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